WILLIAM FORD

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Born: 22nd May 2002, Penrith, Northern England
Nationality: British

PROFILE

PhD candidate in mathematics, jointly between CMAP, École Polytechnique and LMO Paris-Saclay. Wider research interests in the Calculus of Variations, PDE, and applications.

EDUCATION

CMAP École Polytechnique/ LMO Paris-Saclay, PhD in Mathematics *2025 – 2028*

The central goal of my PhD is to better understand the quantitative stability properties of optimal transport. This is crucial for applications - to quantify errors introduced by approximations - as well from a theoretical viewpoint - to better understand the behaviour of the optimal transport problem under deformations of its marginals.

Supervisors*: Dr Michael Goldman, CMAP & Dr Cyril Letrouit, LMO Paris-Saclay.*

IP Paris PhD track/Paris-Saclay M2 Optimisation: 18.1/20 *2024 – 2025*

M2 in theoretical optimisation and links to Partial Differential Equations.

Internship: *Quantitative Stability in Discrete Optimal Transport, CMAP Polytechnique.* *April – August 2025*

Supervisors*: Dr Michael Goldman, CMAP & Dr Cyril Letrouit, LMO Paris-Saclay.*

Durham University MSc Mathematics: First Class – 88/100 (ranked 1st out of 25) *2023 – 2024*

Master specialising in analysis and probability (functional analysis, PDEs, stochastic analysis, ergodic theory).

Research Dissertation: *"Partial Regularity for Optimal Transport Maps Between Uniform Measures"* (86/100)

Supervisor: *Dr Alpár Mészáros, Durham University.*

Durham University BSc Mathematics: First Class – 82/100 (ranked 7th out of 129) *2020 – 2023*

3 year BSc covering a range of topics in fundamental and applied mathematics. Received first class (trés bien) marks in all modules taken and received the Norton Prize in 2021 for outstanding performance.

Research Dissertation: *"Normal Families in Complex Analysis"* (79/100)

Supervisor: *Dr Wilhelm Klingenberg, Durham University.*

Queen Elizabeth Grammar School Penrith *2013 – 2020*

AWARDS AND SCHOLARSHIPS

2024	Sophie Germain M2 Excellence Scholarship, Fondation Mathématique Jacques Hadamard.
2024	PhD track full package M2 + PhD scholarship, Institut Polytechnique de Paris.
2024	Winton MSc Memorial Prize for highest scoring student in Mathematics, Durham University.
2021	Norton Prize for performance in science, St Cuthbert's Society, Durham University.

TEACHING EXPERIENCE

Teaching Assistant | École Polytechnique *2025 – 2026*

How to write mathematics I (one group of 21 students), Numerical Linear Algebra (one group of 20 students)

Teaching Assistant | Durham University *2023 – 2024*

Calculus I and Linear algebra I (two tutorial groups of 12 students)

Mathematics Tutor | MyTutor.co.uk *2021 – 2023*

Tutored GCSE and A-level maths students in both 3 to 1 and 1 to 1 settings.